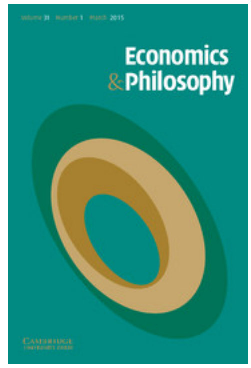

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REAL PATTERNS AND THE ONTOLOGICAL FOUNDATIONS OF MICROECONOMICS

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Most philosophical accounts of the foundations of economics have assumed that economics is intended to be an empirical science concerned with human behaviour, though they have, of course, differed over the extent to which it has been or can be successful as such an enterprise. A prominent source of dissent against this consensus is Alexander Rosenberg. In his recent book, Rosenberg summarizes and completes his statement of a position that he has been developing for some time (Rosenberg, 1992a). He argues that although economists evaluate one another's work according to shared and rigorous standards of adequacy, these standards are strictly internal, like those of mathematics, instead of being derived from discoveries about contingent relationships between theoretical statements and empirical facts. Economics, that is to say, is essentially conceptual rather than empirical inquiry. In the following discussion, I will provide grounds for resisting Rosenberg's conclusion. The point of this criticism, however, is not mainly negative. Most of the historical preoccupations of the philosophy of economics, like those of the philosophy of science in general, have been epistemological. Careful attention to Rosenberg's argument, however, redirects our attention to ontological questions about the *objects* of economics, and this redirection, I will maintain, is to be welcomed. I will argue that while the majority of

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philosophers of economics is correct, as against Rosenberg, in regarding economics as an empirical science, the conventional view as to the identity of its empirical objects should be substantially revised under the pressure of Rosenberg's criticisms. My critical analysis of the implications of Rosenberg's work for the ontological commitments of economics is intended to furnish *one* line of argument towards my own conception of the objects of economics. This conception, which is still preliminary in many respects, will be briefly sketched toward the end of the present paper; a detailed account of it is given in Ross and LaCasse (1993).

One caveat is in order before I proceed. My attention will be largely confined to microeconomics. I make no claim that the position I will sketch provides foundational support for the theoretical and practical structure of macroeconomics. It might appear to be to Rosenberg's advantage that he imposes no such restriction on his analysis, offering it as a general philosophy of economics that both microeconomists and macroeconomists should endorse. In his reach for generality of scope, however, Rosenberg exceeds his grasp. It is among the most apparent and anecdotally well-attested facts about the sociology of economics that microeconomists are frequently uneasy about the foundational integrity of macroeconomics, especially macroeconomics based on models which do not feature optimizing behaviour. Although the rise of rational expectations theory has encouraged, in some quarters, the view that microeconomics will provide foundations for macroeconomics,[1] the fact that many economists themselves do not expect unification should lead us to be wary of presuming this. Prudent regard for the actual dynamics of the discipline makes it wisest to first seek *separate* foundational theories, and *then* ask whether, and/or how, these pieces can be fitted together into a more general interpretation. The following discussion is confined to just one of these three tasks.

The problem of identifying the objects of microeconomics arises from the fact that the most obvious candidates for the job face serious objections. The most natural, and naive, interpretation of microeconomics is to view it as seeking generalizations about the behaviour of actual people under idealized conditions which are most closely approximated in the competitive marketplace. The appeal of the naive view is obvious:

[1] See Lucas, 1973; representative work on real business cycles (Kydland and Prescott, 1982; Long and Plosser, 1983); and recent work on consumption and asset prices. Most of these macroeconomic models directly borrow the tools of microeconomic theory by featuring a 'representative agent' who is infinitely lived and maximizes some utility function. Another unifying view, though one which does not necessarily award pride of place to microeconomic tools, holds that the two branches of economics are foundationally bound by their common goal of analysing economic systems by means of general equilibrium theory. This view seems to be frequently held by economists who study growth, and is sometimes hinted at by Rosenberg himself.

according to it, microeconomics studies people (*qua* rational utility maximizers) in just the way that mechanics studies bodies (*qua* packets of energy subject to limited ensembles of forces), and economics presents no special problems for the philosophy of science. The *defects* of the naive view, however, are just about equally obvious. If the conditions that define 'Rational Economic Man' are treated as idealizations, then we face a battery of well-known problems concerning their empirical motivation and limits of application. In general, a science that traffics in idealizations after the model of classical physics achieves its predictive successes by relaxing the idealizations in particular cases according to principles that can be independently justified in terms of the operating theory. There was a moment in the history of economics, prior to the work of Edgeworth and Pareto, when it might have developed in this way, and thereby have become a branch of the emerging science of psychology. But Pareto's disdain for motivations as 'metaphysical entities' led him to reinterpret utility in a way that foreshadowed Samuelson's later project of showing how to assign preference profiles without making empirical claims or assumptions about individuals other than (apparently) that their demand behaviour could have been generated from a utility maximizing model using *some* fixed set of tastes. I say 'apparently' here because, as many writers have objected, the fact that tastes must be treated as unchanging throughout an analysis, in conjunction with the fact that there is no extra-economic theory of tastes to which economists can defer in relaxing their idealizations, means that data cannot be used as a basis for revising the rationality conditions embedded in a given model; instead, they can only be used to determine whether the model applies at all. But in that case the rationality conditions cannot be idealizations, since idealizations are *abstractions* from empirical regularities, not a priori criteria for theory application. The rationality conditions resemble idealizations with respect to their *function*, which is to permit generalization and analytical tractability, but they do not resemble idealizations with respect to their *justification*. If microeconomics does not describe idealized individuals, then the only way to try to preserve the naive view is to argue that it describes *actual* individuals, that is, that individual people are in fact rational (at least in the marketplace) in the way that revealed preference theory supposes. But that economists are not inclined to this response is suggested by their general indifference to famous experimental results (Nisbett and Ross, 1980; Kahneman *et al*, 1982) suggesting that most people *systematically* violate the assumptions of rational choice theory.

If one concludes that agents are not the objects of empirical study for microeconomics, then one may reasonably wonder whether there are *any* such objects, and this clearly amounts to wondering whether microeconomics is an *empirical* science at all. Accounts of microeconomics as nonempirical may be divided into two broad classes: those that undermine its general epistemic legitimacy, and those that decouple it from empirical pretensions in order to defend it. Proponents of accounts in the first class usually hold economics to be disguised ideology. This is a view which I will not take seriously. While it is no doubt true that many individual economists have been influenced in their work by their personal political persuasions – as is the case in any science – there is no attainable critical perspective from which a (pompous) philosopher could legitimately convict an entire discipline of bad faith. With less malice aforethought, other philosophers have argued that microeconomics is a normative enterprise not in the political sense, but rather according to the sense in which epistemology is a normative enterprise. On this view, microeconomics investigates the *conceptual* structure of maximizing rationality, without pronouncing on the extent to which consumers *are* rational in this way, or ought to be. Just as the epistemologist provides practical advice only for those who already place a high relative value on truth and belief consistency, so the microeconomist's recommendations are conditional on our extrinsically motivated concern with the consistency of acquisitive actions relative to a well defined set of interests. To the extent that one associates this sort of conceptual inquiry with the philosopher's task, microeconomics emerges on this view as a kind of technically specialized philosophy. If it is not an empirical science, it has at least as much epistemic legitimacy as philosophy, which, in the eyes of philosophers, is apt to be plenty.

Attractive though it can be made to seem, this view, I wish to maintain, is false; microeconomics is an empirical science that investigates real phenomena. My claim, to be defensible, must be importantly different from the naive version of it sketched a couple of paragraphs ago, which is hopeless. My alternative is not fully developed; what I will offer instead is a proposal for further investigation. The motivation for this proposal arises from consideration of what is right, and what is wrong, with the view of economics as conceptual inquiry. Here I turn gratefully to Rosenberg, who, as mentioned above, gives us a sophisticated argument to the effect that economics is a legitimate science precisely to the extent that it is properly seen as a conceptual one. Although it will turn out that Rosenberg is wrong, the way in which he is wrong is the best guide we yet have as to what is right.

According to Rosenberg, economics

> is a branch of mathematics, one devoted to examining the formal properties of a set of assumptions about the transitivity of abstract relations: axioms that implicitly define a technical notion of 'rationality', just as geometry examines the formal properties of abstract points and lines (Rosenberg, 1992a, p. 247).

The analogy with geometry is developed at some length in Rosenberg's discussion, and is essential to understanding his justification for claiming that economics is a species of mathematics. Geometry was at one time taken to be an empirical science, whose central terms referred directly to classes of objects in space. The fact that the methods of geometers were uncharacteristic of experimental science, however, generated the problems in the epistemology of geometry that found their most acute expression in Kant's uncomfortable doctrine of the synthetic a priori. Later, the development of non-Euclidean geometries and of relativistic physics demonstrated that geometry is not the empirical science of space, but is instead the study of patterns of relations among abstract concepts defined according to families of axioms. *Part* of geometry's utility stems from the applications that are made of it by physicists, but the epistemic integrity of a given piece of geometric inquiry is independent of the existence of any such uses. The situation of economics, Rosenberg argues, is parallel to this. Economists initially took their central terms to refer directly to agents and to some of their psychological properties. Their indifference to predictive failures, however, and the apparent willingness of early eminences such as Mill to insulate theory indefinitely against the possibility of disconfirmation, led methodologically self-conscious economists to produce a trail of special epistemologies for their discipline, each of which tries implausibly to rationalize the Kantian tension between empiricism and a priorism. (For an outline of this history, see Blaug, 1980, Part II.) As with the epistemology of geometry, we are best freed from such hopeless labours, Rosenberg maintains, when we recognize that the objects of economic inquiry are purely abstract relations. To be sure, economics has applications – principally, whenever we wish to determine the possible range of impacts of some exogenous change in the trading environment of agents evolving in a competitive system, or the possible qualitative influences upon a particular market of a change in the competitive system to which it belongs. That economic arguments usually have the form of possibility proofs, however, should alert us to the fact that they do not directly concern contingent phenomena. Rosenberg concludes that economics, like geometry, is hostage only to standards generated by its own historical dynamic of mathematical rigour. Its authority as a basis for public and personal policy is necessarily limited by this analysis, but, in compensation, systematic predictive imprecision need not threaten the integrity of the discipline.

At least where microeconomics is concerned, the chief merit of this account lies in its explanation of two salient features of the discipline: the fact that greater prestige and significance is accorded to abstract possibility proofs than to elaborations of structural isomorphisms between models and phenomena; and the unusual attitude of economists, by comparison with other scientists, towards the importance of their discipline's empirical track record.[2] I think that the account also has two serious flaws: (1) it depends on a misunderstanding and a misapplication of the distinction between the conceptual and the empirical; and (2) it requires an implausibly strong identification of scientific *theory* with scientific aims and methods. These mistakes abet one another, and so I will seek to substantiate my identification of them together.

Distinguishing conceptual inquiry from empirical inquiry does not require that we revive rigid positivistic distinctions between the a priori and the a posteriori or between the analytic and the synthetic. We could instead agree to count, say, set theory as a paradigm case of conceptual inquiry and terrestrial geology as a paradigm case of empirical inquiry, look for methodological characteristics of the two sciences that seem crucially related to our intuitions, and then sort other investigative enterprises along an intervening spectrum governed by those characteristics. One sorting characteristic which will surely emerge as very important here is the extent to which a discipline treats predictive success as an essential aspect of hypothesis justification.[3] For this reason, Rosenberg focuses much of his attention on the predictive efficacy of economics.

In particular, he relies heavily on the distinction between *quantitative* and *qualitative* prediction. His use of it is inspired by Samuelson's remark that

> In the absence of complete quantitative information concerning our equilibrium conditions, it is hoped to be able to formulate qualitative restrictions on slopes, curvatures, etc., of our equilibrium equations so as to be able to derive definite qualitative restrictions upon the responses of our system to changes in certain parameters. (Samuelson, 1947, p. 20)

Samuelson makes clear in a subsequent passage – which Rosenberg quotes, though missing the point – that his claim is *not* that individual economists who are skilled in the application of theory and closely

[2] The point here is not just the familiar one that economists are less interested in data than most scientists. It is that, in addition, their own view of their discipline's history is uncharacteristic of empirical sciences. Economists do not celebrate epochal instances where theories faced decisive tests – Lavoisier's jar, Halley's comet, the procession of the perihelion of Mercury, etc. Instead, macroeconomists are *embarrassed* by the crudeness of historians of thought who try, for example, to portray the 1929 crash as a turning point in the fortunes of neoclassical theory, and I think that most of them are at some level *relieved* that this apparent 'revolt of the facts' could later be interpreted as cancelled out by the stagflation of the 1970s. I do not mean to be taking a stand here on the epistemic significance of crucial experiments. I take it to be interesting, however, that the mythology of the discipline has so little use for them.

[3] It is possible to dissent from this 'predictive empiricism', as an anonymous referee of an earlier draft of this paper points out. However, Rosenberg clearly accepts it, and so, at least for the purposes of the argument, will I.

acquainted, through experience, with a particular system, cannot often achieve reliable quantitative predictions about the system in question (pp. 26–7). Rather, his contention is that economic *theory* does not by itself provide methodological recipes for quantitative prediction. (I will henceforth refer to 'specific prediction', of which Samuelson's particular notion of 'quantitative prediction' is a special case.) Now, this feature of economics is shared with a number of other thriving sciences, including evolutionary population ecology, cognitive ethology and paleotaxonomy. If this observation is important, it must be because it picks out a class of inquiries for which special epistemological problems arise. *Call* this class a class of 'conceptual' inquiries if you like; but this terminological decision solves no problems unless these sciences relevantly resemble geometry in the way that economics is supposed to. Recall that we are sorting sciences along the conceptual-empirical continuum by reflective equilibrium against our intuitions. My intuitions are pretty sure of themselves in judging that whereas geometry is not about space – even though it can be applied to space – evolutionary population ecology is essentially about the dynamics of populations, cognitive ethology is essentially about animal behaviour, and paleotaxonomy is essentially about evolutionary history and relationships. The point is that it would be a mistake to identify the boundary between conceptual and empirical inquiry with the boundary between sciences whose theories identify strategies for delivering specific predictions and sciences whose theories do not.

Rosenberg does not, exactly, make *this* mistake, since he knows that it would be unacceptable to claim that evolutionary biology is not empirical inquiry. What he seems to suppose instead is that for a science to be empirical it must make predictions that are *relevant* to (successful) specific predictions, whether or not it directly offers such predictions itself (Rosenberg 1992a, p. 70). Evolutionary theory is at least *intended* to be relevant to specific prediction, though Rosenberg appears to be of two minds as to whether it succeeds in this. At one point he implies that to the extent that evolutionary theory relies on adaptationist assumptions, it is *failed* empirical inquiry (pp. 181–5). Later he suggests that evolutionary theory contributes to successful empirical inquiry once it is buttressed by molecular genetics (pp. 233–4). Since Rosenberg does not explicitly summarize his view of the comparison between evolutionary theory and economics, I shall try to do so for him in a way that seems to me to be charitable in the light of what he needs for his larger argument. *If* evolutionary theory is committed to adaptationism, then it is irrelevant to specific prediction, and so is a species either of conceptual inquiry or of failed empirical inquiry.[4] To the extent that it is not committed to

[4] If one takes the view that evolutionary theory is committed to adaptationism, *and* that it is conceptual inquiry, then it might, in fact, be fairly natural to see evolutionary theory as a part of economics!

adaptationism, it is relevant to specific predictions by way of its relationship to molecular genetics. Now, the only science that *could* stand to economics as molecular genetics (presumably) stands to evolutionary theory would be a psychological theory that gave ontological substance to the motivational and intentional states in which microeconomics traffics (pp. 112–51, 235–6). Taking an explicit stand on hot issues in the philosophy of cognitive science, Rosenberg asserts that no such theory is, or will be, forthcoming. Therefore, economics cannot be vindicated as empirical inquiry in the way that evolutionary theory might be; therefore, economics is either failed empirical inquiry or it is conceptual inquiry. This is intended by Rosenberg as an *explanation* of the (putatively) independently established claim that, as a matter of fact, the predictive success rate of neoclassical economics has failed to show improvement over the last century or so (p. 149). From here, he goes on to argue that since our intuitions need not revolt at treating economics as conceptual inquiry, in the way they do where evolutionary theory is concerned, the cross-disciplinary principle of charity that is now a working maxim among philosophers of science should lead us to hypothesize that economics is conceptual inquiry.

There are several tendentious assumptions driving this reasoning. Let us focus first on the relationships between qualitative predictive efficacy, specific predictive efficacy, and the empirical/conceptual dichotomy. Rosenberg provides only a sketchy argument for his judgement that qualitative predictive efficacy is insufficient for successful empirical science, but his basis for it seems to have Lakatosian affinities: he supposes that a scientific research programme which cannot push its predictive power beyond the qualitative must be regarded as degenerating. This invites two immediate criticisms. First, it fails to address the possibility that a science could progressively develop by widening its range of qualitative predictions.[5] Following Dawkins (1986), I believe that this is in fact the principal sense in which evolutionary theory has been progressive. Second, it glosses over the subtlety in Samuelson's distinction, alluded to above, which Rosenberg seems to have missed. One of the principal functions of a theory is to organize and systematize phenomena. A theory that does a good job of this may play an essential role in enabling an individual scientist who is empirically well acquainted with a particular system to provide specific predictions about that system. The extra-theoretical knowledge and practical experience that, along with the theory, constitute necessary conditions for such predictions need *not* always be specifiable as initial conditions by means of parameters recognized explicitly by the theory. The point is that there are two ways

[5] Rosenberg doesn't ignore this possibility. But his sketchy argument against using it as a basis for defending the progressiveness of economics (Rosenberg, 1992a, pp. 106–7) begs the question. More on this below.

in which a theory can meet Rosenberg's criterion of relevance to specific prediction: by spawning a body of specifically predictive theory, or by serving as a conceptual organizer that facilitates a range of low-level quantitative inferences in the company of particular, system-relative empirical facts. Since empirical relevance of this second variety will not (necessarily) generate *theoretical* progress, we might distinguish it from the standard Lakatosian image of progressive science by saying that a science which perpetuates itself by being useful in the second way is *empirically productive.* If philosophers of science have given little notice to empirical productivity, this may reflect a history of undue emphasis on theory as the sole repository of epistemic importance, with which the discipline has been persuasively charged by Hacking (1983) and Cartwright (1983).

The most pressing immediate business of the philosophy of economics, I suggest, is to investigate the question of whether microeconomics and macroeconomics are (separately) empirically progressive in the first sense described above, or empirically productive, or both, or neither. This cannot be accomplished just by conceptual reflection and through surveys of economists' methodological tracts; what we need are close, detailed analyses of particular episodes of economic explanation and model-building, of the kind to which we have become accustomed in the philosophy of physics. My expectation is that such research would find enough clear cases of empirical *productivity* to explain the continuing *survival* – indeed, institutional flourishing – of economics. But my expectation might be wrong, since more overtly sociological, non-epistemic factors could conceivably account for the institutional facts. There is, I think, little to be gained by staging a war of philosophical preconceptions here; we must simply get down to work on cases. On the question of qualitative empirical progressiveness, however, there is more to be said alongside, or in advance of case-work, since the very concept of qualitative empirical progress is unclear, and is implicated in important general issues in the philosophy of science.

One of the chief merits of Rosenberg's work is that it draws our attention away from the mere *extent* to which economic theory can deliver specific predictions, in favour of focusing it on the epistemic dynamics of a field which demonstrably doesn't care very much, at least by comparison with the familiar examples of scientific disciplines, about specific predictions, whether it actually produces them or not. As discussed, I do not find Rosenberg's diagnosis of economics as conceptual inquiry – the cash value of which lies in the appropriateness of his analogy between economics and geometry – to be very convincing. He is driven to this analysis, however, because he simultaneously wants to be charitable towards economics while denying the possibility of a science whose empirical progressiveness can consist solely in expanding the range of its qualitative predictions. This is an important mistake, because it has consequences that ramify through the philosophy of science. Just as it pushes him to an uncomfortable philosophy of economics, it is also implicated in his view of evolutionary theory. Recall that, for Rosenberg, evolutionary theory, despite its qualitative character, is in better shape than economics as an empirical science because, when deployed in conjunction with our understanding of the mechanisms of inheritance supplied by molecular genetics, it is frequently relevant to specific predictions of a kind that it cannot generate on its own. One familiar way of understanding this relevance relation might be as follows: molecular genetics 'cashes out', in some quasi-reductionistic sense, the fundamental causal operators – Mendelian genes – whose *qualitatively* significant properties are described by neo-Darwinian theory; thus neo-Darwinian theory conceptually informs, and perhaps heuristically guides, molecular genetics, but it is molecular genetics, not neo-Darwinian theory, that directly provides specific predictions. This is alright, as far as it goes. But the moment one tries to be *any* more specific than this, Rosenberg's conceptual framework begins to chafe. If the 'cashing out' function alluded to above is explained by holding that classical genetics *really reduces* to molecular genetics, then neo-Darwinian theory will be assimilated to molecular genetics, many of its famous qualitative generalizations will turn out to be false, and we will *not* have a case, after all, of a qualitative theory that is autonomous from the source of specific predictions while being nevertheless relevant to those predictions. On the other hand, if one follows Kitcher (1984) in maintaining that Mendelian genes are irreducible to any types recognized by molecular genetics, and thus that classical genetics is an autonomous special science in the sense of Fodor (1974), then the relationship between neo-Darwinian theory and the specific predictions to which it contributes will be relevant to the former's empirical productivity, but not to its empirical progressiveness. Rosenberg had best take the first approach, lest evolutionary theory turn out to have an epistemic status just like economics after all, in which case he will face the unwelcome choice of either assigning evolutionary theory, with economics, to conceptual inquiry, or acknowledging that there can be qualitative theories that are empirically progressive. Now, the first option is also one which Rosenberg might prefer to resist, since his argument about economics is clearly weakened if it turns out to depend on taking one side of a controversial issue in the philosophy of biology. My suspicion, however, is that Rosenberg's scepticism about qualitatively predictive sciences is ultimately motivated by reductionist intuitions. I will try to substantiate this suspicion, not for the sake of finding error in Rosenberg, but to show how different microeconomics can appear under the light of varying attitudes to reductionism.

When Rosenberg seeks respects in which evolutionary theory is on firmer ground, despite its qualitative character, than economics, his emphasis is revealing. He does *not* cite the notable successes of evolutionary ethologists at predicting and explaining patterns in foraging behaviour, mate selection, reproductive spacing, behavioral thermoregulation, kin preference, territoriality and so on. Instead, in exact contrast to Dawkins' arguments for interpreting the scope of Darwinian theory as broadly as possible, Rosenberg writes as if the sole concern of evolutionary theory were changes in distributions of population characteristics over time, as understood in terms of changes in gene frequencies (Rosenberg, 1992a, pp. 195–6; 1992b, pp. 176–83).[6] One would gather from Rosenberg's remarks that in this area evolutionary theory delivers, and explains, a single class of qualitative results, namely, the tendency of gene frequencies to find and maintain stable equilibria. This, one might suppose, is a rather slight base of accomplishments for a successful science. However, Rosenberg uses it to contrast economics, treated as an empirical science, unfavourably with evolutionary theory (at least with respect to their use of equilibrium concepts; and these concepts, as Hausman (1992) argues, are absolutely fundamental to the structure of economics):

> economic equilibrium theory ... lacks the most important feature that justifies the same kind of thinking in evolutionary biology: independent evidence that there is a stable equilibrium to be explained (Rosenberg, 1992a, p. 195).

This is not very convincing. Both populations and exchange relationships will seem more or less stable depending on the grain of analysis with which they are studied. If someone were surprised that populations are stable to *any* measurable degree, then her surprise could be relieved just by describing natural selection, that is, by reciting Darwin's most basic insight. Similarly, Adam Smith already has the answer to someone who does not understand why exchange relationships are not irremediably chaotic. The question, then, is this: what might justify Rosenberg's view that the focus on equilibria is justified in the case of evolutionary theory in a way that it isn't in the case of neoclassical economic theory? His explicit answer is extremely unsatisfying. Evolutionary adaptational problems, he claims, are to a suitable degree of approximation parametric, whereas

[6] Rosenberg makes clear that he 'do[es] not endorse the view that the theory of natural selection is about gene ratios' (personal communication). This accords with what he says in his detailed study of the foundations of biology (Rosenberg, 1985). But note that I have *not* charged him with holding this view, exactly. Rather, what I read Rosenberg as implying, at least in his briefer treatment of the nature of biology as it relates to economics, is that what makes it possible for evolutionary theory to treat empirically of changes in distributions of population characteristics, despite rendering only qualitative predictions about them, is that these changes are, so to speak, 'ontologically grounded' in measurable changes in gene frequencies.

economic relations are fundamentally nonparametric (pp. 195–6).[7] The first part of this claim is highly questionable. Its acceptance would render mysterious the motivations for, and the excitement generated by, evolutionary game theory, and would undermine some of the most important arguments against Panglossian adaptationism in biology, arguments with which Rosenberg appears to be in sympathy. Rosenberg may, however, have a better answer available to him than this official one – 'available' in the sense that it comports naturally with his stated grounds for doubting that economics is or can be empirically progressive. *If* one supposes that genes are the basic objects of evolutionary theory, and that molecular genetics ostensively, so to speak, substantiates these objects by supplying their detailed mechanics, then one can understand how evolutionary theory can be in good foundational shape as an empirical science even though most of its generalizations are only qualitative. As Rosenberg notes (p. 195), specific predictions are hard to come by in population genetics due to the complexity of the genetic/environmental causal nexus. But there are cases, such as (the philosopher's favourite) that of the heterozygote superiority that keeps human populations in malarial regions polymorphous for the sickle cell allele, where one set of vectors is so causally dominant that local equilibria can be specified, predicted and tested with great accuracy. Finding some such cases in the field, and being able to produce many others in the lab, reassures us that evolutionary theory is tracking real phenomena despite its poor output of important specific predictions. Its failure to be empirically progressive in the conventional sense, then, turns out to be a consequence of the difficulty of gathering data, rather than of conceptual problems in the theory.

My main basis for attributing to Rosenberg the view that it is the molecular substantiation of genes and genetic mechanisms that makes evolutionary theory empirical, in a way that economics can't be, is that he explicitly holds the *lack* of such a substantiator in economics to provide the explanation for the putative failure of economics to satisfy its empirical pretensions (p. 149). If economics is interpreted empirically, then its basic objects, according to Rosenberg, are types of intentions – beliefs, preferences and plans. Microeconomics stands to these entities just as Darwinian theory stands to genes. That is, if we take revealed

[7] Rosenberg does not maintain that evolutionary theory is successful *just* because adaptational games are mainly parametric (personal communication). But this *does* seem to be the basis for his view that biologists can have confidence in equilibrium phenomena in a way that economists can't. It should, in general, be borne in mind that we could do better justice to Rosenberg's views on biology by availing ourselves of his detailed (1985) study of that subject. But the shifts of emphasis between that work and his brief exposition in Rosenberg (1992a) are precisely to be explained, I am suggesting, by appeal to the fact that placing the sort of weight on the qualitative/specific contrast that Rosenberg does in connection with economics complicates and disrupts a sensible philosophy of biology – his own included.

preference theory and von Neumann–Morgenstern expected utility theory as being among the foundations of microeconomics, then microeconomics has nothing to say about what intentions *are*, about their internal dynamics, or about the nature of their realization in agents. It *begins* from whatever assumptions about intentionality are most reasonable, in the light of what we know.[8] The source of improvements in the empirical progressiveness of economics, on this story, must be improvements in what we know about intentions; this is (I here conjecture) supposed to be analogous to the way in which evolutionary theory is made specifically predictively efficacious in principle and, occasionally, in fact, by the increases in our knowledge of genes delivered by molecular genetics. Leaving aside, for the moment, the question of whether this is the right way to view the relationship between neo-Darwinian theory and molecular genetics, note that it is a beguiling line on microeconomics. While microeconomists' rationality conditions *do* function like idealizations on this account, they function as idealizations *should*. The crucial difference between this view and that dismissed earlier as naive, is that according to the former, our idealizations are not about agents but about intentional structures themselves. And, in cognitive science, we *do* have a thriving empirical inquiry into intentional structures – don't we? Can cognitive science play the role for microeconomics that molecular genetics (putatively) plays for neo-Darwinian theory?

Rosenberg's answer to this question is an unequivocal 'no'. Where the philosophy of cognitive science is concerned he is, it develops, a kind of eliminativist. He does not hold, as Paul Churchland has on occasion (e.g., 1979), that intentional attribution and its folk psychological setting will disappear from daily life; but he insists on the part of eliminativism that matters, namely, the claim that there can be no science of propositional attitudes.[9] I will quote Rosenberg at length here, since there are now so many shades of opinion on these matters that the mere assignment of philosophers to isms is not very informative:

[8] Rosenberg explains economists' unshakeable commitment to rationality as arising from the fact that the use of intentional concepts already presupposes *general* rationality, although it can allow for limited departures from any particular set of rationality conditions under specifiable circumstances. I think that he is entirely right about this, though I do not think that it in any way impugns the empirical integrity of the discipline. For reasons which may become clear(er) below, I am unmoved by the arguments of Stich (1990) and others to the effect that no proper science can presume agents to be generically rational.

[9] A referee of an earlier draft points out that Rosenberg's scepticism about the prospects for a science of the intentional can be justified by appeal to Davidson's arguments for anomalism, and thus does not require eliminativism. Note, however, that I am using 'eliminativism' to denote a position that is a bit less radical than Churchland at his most dramatic. By 'eliminativism' I refer just to the thesis that no intentional predicates will figure in a successful scientific theory. While this is weaker than the thesis that the intentional predicates will disappear altogether, it is *stronger* than anomalism, since it is a matter of contention as to whether or not anomalism is compatible with functionalism. (See

> the real source of trouble for the attempt to find *improvable* laws of economic behaviour is something that has only become clear in the philosophy of psychology's attempts to understand the intentional variables ... '[B]eliefs' and 'desires' ... do not describe 'natural kinds'. They do not divide nature at the joints. They do not label types of discrete states that share the same manageably small set of causes and effects and so cannot be brought together in causal generalizations that improve on our ordinary level of prediction ... let alone attain the sort of continuing improvement characteristic of science. We cannot expect to improve our intentional explanations of action beyond their present levels of predictive power. But the level of predictive power of our intentional theory is no higher than Plato's. (p. 235)

To the extent that Rosenberg's argument depends on eliminativism, it is seriously weakened. Far from being a consensual position in the philosophy of cognitive science, as the first sentence quoted above suggests, eliminativism is a radical, minority view. This is not the place to debate the merits of eliminativism; of more interest in the present context is Rosenberg's *route* to eliminativism:

> The more we learn about the way the brain stores information of the sort to be represented in our beliefs and desires, the more it becomes apparent that there is no fact of the matter about exactly what we want and exactly what we believe. That is, the propositional content of these states is misleadingly precise in the preferences and expectations it attributes to us. (p. 129)

While most philosophers, as noted, are not eliminativists, many philosophers agree with Rosenberg that the context-sensitivity of intentional ascription, and the accompanying 'fuzziness' of intentional entities, leads toward eliminativism; this, combined with the widespread conviction that eliminativism is nevertheless preposterous, is what causes philosophers such as Fodor (1987) to hold out under increasing strain for classical computationalist theories of cognition. However, Dennett (1991), in an important paper, argues that eliminativists and their classical computationalist arch-foes share a common assumption that fuels the above debate: an undermotivated brand of reductionism. Following Dennett's case against this reductionism will enable us to understand Rosenberg's most interesting mistake, and, in the process, will point the way toward an alternative conception of the foundations of microeconomics.

Lycan, 1988, Part 1, for an extended discussion.) Rosenberg's explanation of the failure of economics as an empirical science cannot allow for the possibility of a functionalist theory of mind, so he *needs* something stronger than anomalism. As the referee notes, what Rosenberg actually says in his new book *is* in fact, more suggestive of eliminativism, as I am using the term.

Dennett's argument is difficult to summarize, because its structure is entirely inductive. Through reflection on a number of cases, he invites us to recognize that our scientific explanatory practices abound with uses of and references to irreducible, abstract *patterns*. Philosophers' attitudes toward the ontological status of such patterns have often been highly inconclusive, not to say shifty. Consider rates of inflation (my example, not Dennett's). Are rates of inflation real? Well, they're not *fictitious;* they're not unreal in the way that Tarzan is unreal. A popular position these days is to say that rates of inflation *supervene* upon unnameable armies of micro-events that constitute the true bedrock of reality. This view revives, in fancy dress, Aristotle's belief in *gradations* of reality; rates of inflation are real, but not *as* real as, for example, the notice on Shopkeeper Smith's front window announcing a 10% price rise. The intuitions that support such a metaphysic are as old as philosophy, and Dennett is obviously in no position to *refute* them. He can be read as forcefully pointing out, however, that they do not comport easily with the prevailing pragmatism about ontology that is almost everybody's official position, according to which ontological commitments are determined by our scientific practices, rather than the other way around. In any case, Dennett's main concern is not to argue directly against reductionism, but to assuage worries that an extreme ontological liberalism that sanctions abstract patterns as readily as it does micro-particulars necessarily breeds instrumentalism, irrealism or idealism. Fears on this score are motivated by the fact that patterns are perspective-dependent. What Dennett shows is that ontologists can take perspectives seriously without abandoning realism, so long as they have well motivated epistemic criteria for distinguishing between, on the one hand, patterns which, once recognized from a perspective, serve irreplaceable explanatory and predictive purposes, and, on the other, patterns which are arbitrarily cobbled together under idiosyncratic perspectives to no general epistemic gain.

It is not possible, in this space, to say enough to convince the reader that Dennett is right. I will instead just summarize his conclusion, while offering a reminder of its most basic motivation: since atomistic reductionism seems to be a gratuitously strong metaphysical view, and the currently popular abandonment of ontology urged by Rorty (1979) to be an over-reaction to it, some attempt at a compromise position is in order. Here, then, is Dennett's. To be a real pattern is to be a type of relatively stable, enduring structure that encodes information in a compressed form about other types of structures such that that information can be registered by some physically possible perceiver. More succinctly, what makes something real is both its stability over time under a perspective and its utility as a carrier of information. Two aspects of this position prevent it from sliding into idealism or constructivism: (1) what determines the range of physically possible perceivers is itself the non-constructed, mind-independent structure of the world; and (2) questions about which structures encode information about which others, and about whether a given encoding is more compressed than a bit-map representation and, hence, epistemically useful in Dennett's required sense, can be investigated by the usual objective methods of science. To be sure, these ontological criteria are extremely liberal; according to Dennett, the world is teeming with real patterns. But, if she is to avoid begging the question by appealing to reductionistic intuitions, then an opponent owes us an argument as to what is wrong with ontological liberalism. (Note that, thanks to the epistemic utility criterion, Dennett's view does *not* violate Ockham's principle.)

Dennett's own main use of his ontological thesis is against the argument that goes from the denial of Fodorian micro-realism about propositional attitude states to eliminativism. That cognitive science has failed to find neurophysiological or discrete computational syntactical bearers of propositional content does not imply that the ontology of folk psychology is mistaken, or that it cannot serve as the basis for scientific elaboration, for folk psychology, according to Dennett, has (imperfectly, to be sure) locked onto a set of real patterns. Dennett's conclusion thus directly undermines Rosenberg's explanation of the putative empirical failure of economics. Its significance in that context runs deeper, however. As I have been at pains to show, Rosenberg is consistent in his views of the several sciences that fall under his comparative gaze in the course of his work on economics. His denial of the possibility of purely qualitative empirical progress, his nuanced view of the empirical status of neo-Darwinian theory, and his denial of empirical content to microeconomic theory can all be explained, I suggest, as resulting from his failure to acknowledge the reality of abstract patterns. If there are no real patterns, but only, in the end, real particulars, then theories which do not go beyond qualitative prediction are theories which are imprecise and incomplete, which fail, in an important sense, to be *accurate*. Empirical progress must depend on *empirical* contact with the world; it must involve showing that the theory in question says an increasing proportion of true things about real entities. Thus the empirical content of neo-Darwinian theory can be saved *if*, contrary to Kitcher (1984), its central entities are given 'substance' by molecular genetics. Thus, too, economics could be an empirical science only if there were real entities for it to be *about*. The thesis that it is about agents is, as we were reminded at the beginning of the paper, a non-starter. A more promising suggestion is that it is about the role played by intentional structures in strategic interaction. But these structures are not yet themselves bona fide empirical objects, in Rosenberg's view. If economics is to have an empirical subject matter, then its putative objects must be substantiated by cognitive science. At this point, the general failure to acknowledge real patterns arises again in this more particular context to suggest that cognitive science *doesn't* substantiate intentional structures. And thus we arrive at Rosenberg's conclusion: either we must push the analogy between economics and geometry, implausible though it may be, in order that economics turns out to be respectable conceptual inquiry, or we must consign economics to the realms of pseudo-science and propaganda.[10]

When matters are presented in this light, it emerges that the emphasis on prediction was something of a distraction. Emphasis shifts from epistemology to ontology: an empirical science needs empirical objects of study, and it is not obvious that economics has any. Since this was the point made at the beginning of the present paper, it may seem that our critical progress through Rosenberg has just led us around a circle. I think that something has been accomplished, however. I would argue, though I cannot do so here, that the main controversies in the philosophy of economics have *always* been driven, below the surface, by difficulties in identifying the subject matter of economics. The nature of Rosenberg's mistake now serves to show that we have consistently been looking in the wrong place. Having failed to discharge an unwarranted metaphysical assumption that abstract patterns cannot be empirical objects, most inquirers have searched along the ground, assuming that the objects of economics must be 'found' objects, that they must arise from a framework extrinsic to the discipline – from our natural ontology, in the cases where individuals and intentions have been proposed as the relevant objects. An alternative view, to the effect that economic theory *creates* a perspective from which real patterns emerge that would otherwise be invisible, has not been seriously considered. Is such an alternative credible?

Among the philosophers and economic methodologists who have focused most explicitly on the question of the *objects* of economic study is Sir John Hicks (1939). As Rosenberg (1980, p. 87) notes, Hicks broke with his tradition in asserting that economics is not about individual behaviour. Of more interest in the present context, his proposed alternative objects were *abstracta par excellence*: economics, he declared, is concerned with the properties of markets and whole economies. In his view, microeconomics is just a sort of calculus that serves macroeconomics, and the former's referentia – agents, preferences, etc. – are essentially theoretical *fictions*. Now, this is not a position which I wish in any way to endorse. If macroeconomics is a sound empirical science, then the phenomena that it tracks are surely abstract real patterns; but, as

[10] I should make clear that this is *not* intended as a *reconstruction* of Rosenberg's argument. It is supposed to be a diagnosis of a mistaken view of the relationship between theory justification and ontological commitment which, if attributed to Rosenberg, explains at a single stroke the various, more particular, mistakes that I have discussed. The main point of the discussion of Rosenberg's eliminativism was to provide additional evidence licensing this attribution.

mentioned earlier, I wish to remain strictly neutral on the question of whether macroeconomics and microeconomics have their detailed foundations in common, and I certainly do not suppose that microeconomics derives its justification from its service to macroeconomic ambitions. What is interesting about Hicks is that when he poses a direct question about ontology in connection with economics, and when, as in his account, this question is not complicated by philosophical sophistication concerning explanation and prediction, he sees, almost at a glance, that economists are concerned with abstract patterns. It is important to note that these patterns could not themselves have been detected prior to the development of some economic theory. No one in ancient Athens was in a position to know that there were rates of inflation and various local equilibria of demand and supply. An economic *perspective* had first to be found, from which the information carried in such patterns could be recognized. Roman and medieval thinkers went a considerable way towards developing such a perspective, and the greatest single contribution to its clarification belongs to Adam Smith. But the early economists did not *construct* economic phenomena; there *were* rates of inflation in ancient Athens, which can sometimes be determined in retrospect and which can serve in historical explanations.[11] Economic patterns are only evident from an economic perspective, and the more sophisticated the perspective, the more patterns will turn up. But they are *real* patterns nevertheless.

So what about microeconomics? Which real patterns are its objects of study? To begin with, the best *general* description of the subject matter of microeconomics remains Robbins's classic definition of it as 'the science which studies the relationship between ends and scarce means which have alternative uses' (Robbins, 1932, p. 15). That there *is* such a relationship – that there is a real pattern here to be investigated – is the most basic insight of microeconomics. Study of this relationship reveals further, deeper patterns: various abstract dilemmas, strategic options and motivations for competitive responses reoccur under broad ranges of particular conditions. The most appropriate name for these abstract phenomena is 'games'. This proposed use of the term is broader than that

[11] An example which helps to make the point involves the Roman emperor Diocletian (284–305 AD). He found himself confronted by ruinous inflation that had been triggered by generations of extortion on the part of imperial soldiers, who routinely demanded pay rises as bribes against fomenting mutiny. The standard response of the imperial administration had been to meet the demands by minting additional coin. Diocletian recognized that there *were* inflationary phenomena, in the sense that he saw that money was declining in value because of continual increases in prices. But he and his advisors had no grasp of the relevant structural relations. Not understanding that failure to control the output of the treasury was the cause of the inflation, they fixed prices by edict. As any economist would predict, this caused goods to disappear from the market, and resulted in famine.

which informs game theory, but it is not in conflict with it. Game theory, we may say, studies the most interesting class of games, namely, those in which actions by any one player can directly affect the payoffs of the others, and where this interdependence is recognized by the agents. The decision problems on which neoclassical theory sharpened its analytical tools can be represented as games in which any one player's influence on the others is not measurable because the number of agents is large, and where one player has a stake in the coordination of the market outcome (as in the classic example of the auctioneer hammering down the Walrasian equilibrium). My claim, then, in summary form, is that microeconomics is the study of games, which are real patterns.

As indicated earlier, I do not have space here to provide a full defense of this claim. It is apt to seem intolerably unclear and undermotivated, however, in the absence of at least one example. Let me, then, briefly provide one. The basic idea here is that the games depicted in abstract microeconomic models[12] are *types* of real patterns, which must be tracked if their tokens are to be properly explained and predicted. Consider, then, as an example of such a type, the game that is well known throughout the game theory literature in economics as 'Matching Pennies'. It takes its name from its most familiar toy instantiation. Imagine that there are two players, who may not communicate with each other. Each independently places a coin, showing either heads or tails. The payoff matrix is shown in Figure 1.

		Player I	
		HEADS	TAILS
Player II	HEADS	(2, 2)	(0, 0)
	TAILS	(0, 0)	(1,1)

FIGURE 1

By 'Matching Pennies', I will henceforth refer to the class of games that share two features of this toy instantiation: there are multiple equilibria, and some of them can be Pareto-ranked. Discovering that the 'Matching Pennies' pattern is instantiated in a strategic interaction can thus provide the essential basis for predicting that the interaction may

[12] What counts as a 'microeconomic model', as I here intend the term, is determined by properties of the model, not by the subject matter with which it is concerned. So models built by biologists doing evolutionary game theory are microeconomic models.

yield outcomes that seem to be Pareto-inferior, and for explaining Pareto-inferior outcomes should these arise. The mainstream microeconomic literature features a number of cases of such explanation (though of types, rather than tokens, of interactions; these types, too, are real patterns). In a repeated game where a few firms meet each other in a market for a homogenous good, any feasible payoff can be attained as a subgame perfect Nash equilibrium. Thus there are multiple equilibria and a subset of these can be ranked; this is a case of Matching Pennies.[13] Similarly, Howitt and McAfee (1992) show that in a situation where workers search for employment and firms search for employees, and where it is optimal for both sets of agents to match their intensities of search, there are two equilibria. Since one equilibrium features a higher level of unemployment than the other, they are Pareto-ranked. Matching Pennies is thus instantiated again.

While examples such as these are the most interesting, citing them *alone* in the present context will not suffice, since seeing the modelled types of interactions as empirical phenomena *already* rests on an appeal to the real patterns metaphysic. Let me, then, describe what I take to be a token whose *individuation* requires no theoretical sophistication, but which cannot be fully understood unless it is recognized as a token of Matching Pennies. Since one of the epistemically important characteristics of many real patterns is robustness across conceptual domains, it is helpful in any case to find an example which falls outside of the classical subject matter of economics. Here, then, is an application drawn from my own doorstep. In the city of Ottawa, where I live, an unusually high proportion of drivers from out of town get into accidents at intersections. Anecdote suggests that they show marked impatience when behind local drivers at a light that has just turned green; the locals evidently seem to them to be sluggish in getting off the mark. Ottawa motorists might be happy to explain these facts by supposing that the visitors are reckless, impatient drivers by comparison with themselves. But one other observed fact, in conjunction with the recognition that this is a case of Matching Pennies, suggests an analysis that is less flattering to the locals. The additional fact is that Ottawa drivers *never* stop on yellow lights; instead, they speed up to race through them. Now, consider this as a game of Matching Pennies. Suppose the agents in the game are drivers travelling in perpendicular directions with respect to each other through an intersection whose light is changing. Hoople is the last driver through the intersection in his direction; he can either speed up or stop on yellow. Ziffel is the first driver through the intersection in the other direction; he can delay before going on the green, or proceed at once. The two

[13] This is implied, for the infinitely repeated game with discounting, by results due to Fudenberg and Maskin (1986), and, for quantity-picking oligopolies in the case of finitely repeated games, by results due to Benoit and Krishna (1985).

equilibria in this game tell us that, for both the case involving only Hoople and Ziffel, and the general case involving many drivers, there are two stable situations, namely, that in which everyone speeds through yellow and all opposing drivers wait, or that in which everyone stops on yellow and all opposing drivers proceed. Ottawa drivers have converged on the first equilibrium, and this causes difficulties for visitors whose communities have converged on the other. Note also that the Ottawa equilibrium permits fewer cars to turn left on each light than does the other. Since this contributes to gridlock, which inconveniences everyone, it is reasonable to claim that the traffic pattern in Ottawa has locked onto a Pareto-inferior equilibrium. Understanding the pattern as a case of Matching Pennies explains this.

Microeconomists will be familiar with many analyses of this sort. My point in running through one was to provide an 'intuition pump' (to borrow a phrase from Dennett) for my main proposal. One does not fully understand the unusual traffic behaviour in Ottawa unless one sees it as an instance of Matching Pennies (though, of course, to do this one need neither conceptualize nor label it using precisely the notation of game theory). Matching Pennies thus satisfies the criteria for being a real pattern. In this respect, it is a typical microeconomic object.

Again, I stress that the purpose of the example is to *clarify* my proposal, and not to justify it. In the same motivational spirit, I will conclude by reviewing its affinities and differences with Rosenberg's view. Like Rosenberg, I claim that microeconomists study abstract relationships between elements that are specified by the axioms of their own basic theoretical structures. That Rosenberg and I agree on this much is a good thing; something would be seriously wrong if we were at odds in our capacities as 'journalists' over what microeconomists, as a matter of fact, do with their time. But this raises the worry that we *dis*agree only over what metaphysical spin to attach to our common description of microeconomic practice: I claim that the relationships studied by microeconomists are empirical phenomena, whereas Rosenberg claims that they're conceptual phenomena. Philosophers of science of a strongly empiricist bent, who believe that metaphysical annexes to scientific inquiries are epistemically gratuitous, may thus be inclined to dismiss the issues with which Rosenberg and I are concerned as pseudo-problems. I contend, however, that there will be significant differences over what one takes to be the most interesting problems and the most important tasks for microeconomics, depending on whether one prefers Rosenberg's analysis or mine. On my account, microeconomists should be expected to discover new games, new classes of games, and new properties of games *partly* by pure mathematical deduction, but mostly by reflection on the sorts of strategic situations, as they arise for real agents, that constitute empirical tokens of games. Thus, I take Becker's applications of microeconomic models to non-market phenomena, such as marriage and discrimination (Becker, 1957, 1975, 1976), and the fruitful interanimation of microeconomics and evolutionary theory launched by the work of Maynard Smith (1982),[14] to be among the most exciting exemplars of empirical progress in microeconomics. By contrast, Rosenberg views Becker's work, at least, as resting on a philosophical muddle, that is, as forcing contingent, essentially historically conditioned phenomena into a narrow conceptual space which there is no good reason for expecting them to fit (Rosenberg, 1979).[15] According to Rosenberg, microeconomists will be best off if they recognize that they are not empirical scientists and are thus freed from the constraints of empirical science to settle into happy lives as deductive inquirers exploring an arcane but fascinating branch of mathematics. (It should be added that, on Rosenberg's account, they also get to contribute to the equally non-empirical investigations of normative political philosophers working in the contractarian tradition [Rosenberg, 1992a, 215–27]). As I have it, by contrast, microeconomists are right to regard themselves as investigating properties of the world, and their current confusion over *which* elements of the world they are responsible for is to be explained by the fact that most of them, like most philosophers, haven't yet had time to acquire and assimilate Dennett's insights concerning real patterns.

My proposal as to the philosophical foundations of microeconomics is, most importantly, intended to motivate a research programme. The philosopher of science who seeks to elucidate the foundations of microeconomics, and whose working methodology involves presuming that there *are* sound foundations to be elucidated, should first identify the real patterns with which microeconomists are concerned. She should then try to explain, with reference to particular models and applications, how microeconomic methods give epistemic access to those patterns. This will often involve her in ontological debates over which classifications of games and types of games best respect the sometimes competing demands of empirical adequacy and conceptual consistency. She will try to settle such issues by working to reflective equilibrium between her basic intuitions, as honed by philosophical practice, her understanding of the explanatory practices of microeconomists, and the data about well established classes of games that microeconomists unearth. This descrip-

[14] Maynard Smith's own work, and that of the evolutionary game theorists who immediately followed him, was originally fruitful for biologists, rather than for economists. But recent results relating the formal equilibrium concepts of evolutionary game theory to those of conventional game theory promise exciting new applications of the former to problems in microeconomics. (See Mailath, 1992; Young, 1993a, 1993b.) In particular, the sort of attention to the *history* of games made possible by evolutionary game theory offers some promise of taking the sting out of the folk theorem.

[15] Rosenberg has other reasons, rooted in his distrust of adaptationism in biology, that should lead him to take a sceptical attitude toward evolutionary game theory.

tion of a research strategy will sound familiar to mainstream philosophers of science who have dealt mainly with physics or biology. This should further recommend my approach to those who continue to hope, despite many decades of philosophical scepticism, that microeconomics can be shown to have a place among the sound empirical sciences.

REFERENCES

Becker, G. 1957. *The Economics of Discrimination*. University of Chicago Press

Becker, G. 1975. *Human Capital*. University of Chicago Press

Becker, G. 1976. *The Economic Approach to Human Behaviour*. University of Chicago Press

Benoit, J., and V. Krishna. 1985, 'Finitely repeated games.' *Econometrica*, 53:305–22

Blaug, M. 1980. *The Methodology of Economics*. Cambridge University Press

Cartwright, N. 1983. *How the Laws of Physics Lie*. Oxford University Press

Churchland, P. M. 1979. *Scientific Realism and the Plasticity of Mind*. Cambridge University Press

Dawkins, R. 1986. *The Blind Watchmaker*. Longman's

Dennett, D. 1991. 'Real patterns.' *The Journal of Philosophy*, 88:27–51

Fodor, J. 1974. 'Special sciences, or the disunity of science as a working hypothesis.' *Synthese*, 28:77–115

Fodor, J. 1987. *Psychosemantics*. MIT Press/Bradford

Fudenberg, D., and E. Maskin. 1986. 'The folk theorem in repeated games with discounting or with incomplete information.' *Econometrica*, 54:533–54

Hacking, I. 1983. *Representing and Intervening*. Cambridge University Press

Hausman, D. 1992. *The Inexact and Separate Science of Economics*. Cambridge University Press.

Hicks, Sir J. 1939. *Value and Capital*. Oxford University Press

Howitt, P. and R.P McAfee. 1992. 'Animal spirits.' *American Economic Review*, 82:493–507

Kahneman, D., P. Slovic and A. Tversky. 1982. *Judgement Under Uncertainty: Heuristics and Biases*. Cambridge University Press

Kitcher, P. 1984. '1953 and all that: a tale of two sciences.' *The Philosophical Review*, 93:335–73

Kydland, F. and E. Prescott. 1982. 'Time to build and aggregate fluctuations.' *Econometrica*, 50:1345–70

Long, J. and C. Plosser. 1983. 'Real business cycles.' *Journal of Political Economy*, 91:39–69

Lucas, R. 1973. 'Some international evidence on output-inflation tradeoffs.' *American Economic Review*, 63:326–34

Lycan, W. 1988. *Judgement and Justification*. Cambridge University Press

Mailath, G. 1992. 'Introduction: symposium on evolutionary game theory.' *Journal of Economic Theory*, 57:259–77

Maynard Smith, J. 1982. *Evolution and the Theory of Games*. Cambridge University Press

Nisbett, R. and L. Ross. 1980. *Human Inference: Strategies and Shortcomings of Social Judgement*. Prentice-Hall

Robbins, L. 1932. *An Essay on the Nature and Significance of Economic Science*. Macmillan

Rosenberg, A. 1979. 'Can economic theory explain everything?' *Philosophy of the Social Sciences*, 9:509–29

Rosenberg, A. 1980. 'A sceptical history of microeconomic theory.' *Theory and Decision*, 12:79–93

Rosenberg, A. 1985. *The Structure of Biological Science*. Cambridge University Press

Rosenberg, A. 1992a. *Economics: Mathematical Politics or Science of Diminishing Returns?* University of Chicago Press

Rosenberg, A. 1992b. 'Neo-classical economics and evolutionary theory: strange bedfellows?'

In *Proceedings of the Philosophy of Science Association 1992*, Vol. 1, pp. 174–83. D. Hull, M. Forbes and K. Okruhlik, (eds.).

Ross, D. and C. LaCasse. 1993. 'Toward a new philosophy of positive economics'. Working Paper No. 9324, Groupe de Recherche en Épistémologie Comparée, Université du Québec à Montréal; Working Paper No. 9319E, Department of Economics, University of Ottawa

Rorty, R. 1979. *Philosophy and the Mirror of Nature*. Princeton University Press

Samuelson, P. 1947. *Foundations of Economic Analysis*. MIT Press

Stich, S. 1990. *The Fragmentation of Reason*. MIT Press/Bradford

Young, H. 1993a. 'The evolution of conventions.' *Econometrica*, 61:57–84

Young, H. 1993b, 'An evolutionary model of bargaining.' *Journal of Economic Theory*, 59:145–68.